CHEMISTRY
Exam + revision pack

GRADE 11 | UNITS 1 AND 2

Prepared from your shared revision booklet, Chapter 1 quiz, lesson PPTs and photographed homework. The format follows those materials. The teacher has not supplied the actual exam; the selection, marks and 60-minute timing below are practice assumptions.

Your fastest route tonight

1. Learn the rules and worked examples in the revision section.

2. Attempt the 60-mark paper with the answers covered. Allow about 60 minutes.

3. Mark it, then practise the questions you missed in the game.

4. Before bed and tomorrow, recall the definitions and redo missed calculations without notes.

Most strongly supported question types

Particle counts; isotope weighted averages; atomic models; definitions; periodic comparisons with reasons; ionic formulas and names; molecular prefixes; Lewis structures and lone pairs; conductivity and bond polarity.

Also revise

Radiation and isotope uses, flame tests, scientific knowledge, green chemistry, and sweetener-selection criteria. These appear in the lesson material even when less prominent in the quiz.

Notation

Plain-text formulas use digits as subscripts: Al2O3 means two aluminium and three oxygen atoms. A charge such as SO4^2- is the overall ion charge. A dot in CoCl2.6H2O separates water of crystallization.

Predicted exam | 60 marks

Name: _____ Date: _____ Suggested time: 60 minutes

Use a periodic table and calculator for practice. Check your teacher's permitted equipment for the real exam. Show working and units. Draw Lewis diagrams in pencil.

A | Multiple choice

Choose one answer for each question. 8 marks.

1. Atomic number is the number of: A neutrons; B protons; C protons + neutrons; D shells. [1 marks]

2. Which particle is negative? A proton; B neutron; C electron; D nucleus. [1 marks]

3. Across a period, first ionization energy generally: A increases; B decreases; C stays constant; D becomes zero. [1 marks]

4. Which scientist left gaps while arranging elements mainly by atomic mass? A Bohr; B Moseley; C Mendeleev; D Chadwick. [1 marks]

5. Which conducts when molten but not as a solid? A CO_2; B KCl; C CH_4; D O_2. [1 marks]

6. Which is polyatomic? A Na^+; B Cl^-; C NH_4^+; D Mg^{2+}. [1 marks]

7. Which bond is non-polar? A H-Cl; B N-N; C O-H; D C-O. [1 marks]

8. A lilac flame suggests: A Na^+; B K^+; C Li^+; D Cu^{2+}. [1 marks]

B | Definitions

Give concise definitions with the key scientific words. 8 marks.

9. Define isotope. [2 marks]

10. Define first ionization energy. [2 marks]

11. Define electronegativity. [2 marks]

12. Define ionic bond. [2 marks]

C | Calculations

Show your working. 7 marks.

13. A particle has 17 protons, 18 neutrons and 18 electrons. Find Z, A and its charge. [3 marks]

14. Isotopes have masses 10 u (20%), 11 u (30%) and 12 u (50%). Calculate the average atomic mass and explain why it is not a whole number. [4 marks]

D | Order and explain

Give the order or comparison AND the reason. 6 marks.

15. Arrange F, Cl, Br and I by increasing atomic radius. Explain. [2 marks]

16. Which is larger, Na or Na+? Explain. [2 marks]

17. Why does atomic radius generally decrease from left to right across a period? [2 marks]

E | Formulas and names

One mark per correct formula/name. 10 marks.

18. Write formulas: (a) aluminium oxide; (b) magnesium phosphate; (c) iron(III) chloride; (d) ammonium sulfate; (e) carbon tetrachloride. [5 marks]

19. Name: (a) FeS; (b) Mg(ClO3)2; (c) PCl5; (d) N2O5; (e) CoCl2.6H2O. [5 marks]

F | Draw Lewis structures

Show all lone pairs and any overall charge. 9 marks.

20. Draw NH3, H2O and one resonance contributor for CO3^2-. [9 marks]

G | Explain and apply

Use complete scientific explanations. 12 marks.

21. Explain why solid KBr does not conduct but molten KBr does. [4 marks]

22. Compare how ionic and covalent bonds form. [4 marks]

23. EN(C)=2.6 and EN(O)=3.4. Find the difference and classify the C-O bond. [2 marks]

24. Identify (a) radiation consisting of two protons and two neutrons; (b) radiation consisting of high-energy electromagnetic waves. [2 marks]

Marking guide | Sections A-D

A | Multiple choice

1. B. Proton number identifies the element.

2. C. An electron has charge -1 and is outside the nucleus.

3. A. Greater effective nuclear attraction generally makes electron removal harder.

4. C. Modern ordering uses atomic number.

5. B. Molten KCl has mobile ions; solid KCl has fixed ions.

6. C. Ammonium contains several bonded atoms with overall charge +1.

7. B. Identical nitrogen atoms share equally.

8. B. Potassium produces the lilac flame in the course examples.

B | Definitions

9. Atoms of the same element (same proton number) [1] with different neutron numbers [1].

10. Energy required to remove one electron [1] from an isolated gaseous atom [1].

11. Ability of a bonded atom [1] to attract shared/bonding electrons [1].

12. Electrostatic attraction [1] between oppositely charged ions [1].

C | Calculations

13. Z=17 [1]; A=17+18=35 [1]; charge=17-18=-1 [1]. It is a chloride ion.

14. Fractions 0.20, 0.30, 0.50 [1]; weighted expression 10(0.20)+11(0.30)+12(0.50) [1]; 11.3 u [1]. It is a weighted average of different isotopes, not the mass number of a single atom [1].

D | Order and explain

15. F < Cl < Br < I [1]. Down the group, extra occupied electron shells increase radius [1].

16. Na [1]. Formation of Na+ removes its outer electron and occupied outer shell; remaining electrons are more tightly held [1].

17. Nuclear charge/proton number increases [1] while added electrons occupy the same main shell, so stronger effective attraction pulls them closer [1].

Marking guide | Sections E-G

E | Formulas and names

18. (a) Al2O3; (b) Mg3(PO4)2; (c) FeCl3; (d) (NH4)2SO4; (e) CCl4. One mark each. Balance charge for ionic substances; use prefixes for molecular compounds.

19. (a) iron(II) sulfide; (b) magnesium chlorate; (c) phosphorus pentachloride; (d) dinitrogen pentoxide; (e) cobalt(II) chloride hexahydrate. One mark each.

F | Draw Lewis structures

20. NH3: three N-H bonds [1], one N lone pair [1], correct total 8 electrons/complete H duets and N octet [1]. H2O: two O-H bonds [1], two O lone pairs [1], total 8 electrons and correct octet/duets [1]. CO3^2-: central C with one double and two single C-O bonds [1], correct O lone pairs (2 on double-bonded O; 3 on each single-bonded O) [1], brackets and overall 2- charge [1]. Total 24 electrons. See diagrams in the revision section.

G | Explain and apply

21. Solid: ions exist [1] but are fixed in a lattice/not free to move [1]. Molten: ions are mobile [1] and carry charge through the substance [1]. Do not say there are no ions in the solid.

22. Ionic formation involves electron transfer [1], producing positive and negative ions held by electrostatic attraction [1]. Covalent bonding involves sharing [1] electron pairs between atoms [1].

23. Difference=|3.4-2.6|=0.8 [1]; polar covalent under the supplied course convention [1]. Oxygen is partially negative.

24. (a) alpha [1]; (b) gamma [1].

Revision | Learn, cover, recall

A correct answer from memory is more useful than recognizing an answer on the page. Study each box, cover it, and explain the rule with an example.

Particles and ions

Z = protons. A = protons + neutrons. Neutrons = A - Z. Charge = protons - electrons. In a neutral atom, electrons = protons.

Protons: +1, nucleus. Neutrons: 0, nucleus. Electrons: -1, outside nucleus, much lighter. Example: 17 p, 18 n, 18 e gives Z=17, A=35, charge -1. Changing electrons makes an ion; changing protons changes the element.

Isotopes and averages

Isotopes: same proton number, different neutron numbers. Average atomic mass = sum of (isotope mass x abundance/100).

Example: 35 u at 75% and 37 u at 25% gives 35(0.75)+37(0.25)=35.5 u. Check the answer lies between the isotope masses and nearer the more abundant isotope. Atomic mass unit: 1/12 of the mass of a carbon-12 atom.

Models and electron shells

Dalton: solid-sphere model. Thomson: electron, plum-pudding model. Rutherford: small dense positive nucleus, mostly empty space. Bohr: definite energy levels. Chadwick: neutron.

Most alpha particles passed through gold foil; a few deflected strongly. That supports Rutherford's conclusions. First-20-element shell examples: Al 2,8,3; S 2,8,6; Cl 2,8,7; K 2,8,8,1; Ca 2,8,8,2. Light is emitted when excited electrons fall to lower levels.

Revision | Continued

Periodic table and families

Modern table: increasing atomic number. Mendeleev used atomic mass and left gaps. Group = vertical column; period = horizontal row. Main-group family similarities come from similar valence electrons.

Group 1 alkali metals: 1 valence electron. Group 2 alkaline earths: 2. Group 17 halogens: 7. Group 18 noble gases: full shell; helium has 2. Metals conduct and are usually lustrous; metalloids have mixed properties; non-metals usually conduct poorly.

Trends: explain the reason

Atomic radius increases DOWN a group (more shells and shielding), decreases ACROSS a period (greater nuclear attraction in the same shell). Ionization energy and electronegativity generally follow the opposite pattern.

First ionization energy: energy to remove an electron from an isolated gaseous atom. Electron affinity: energy change when a gaseous neutral atom gains an electron; many exceptions occur. Cations are smaller than their atoms; anions are larger. F < Cl < Br < I in radius; Li < Na < K < Rb.

Ion charges and the zero-sum rule

Cations lose electrons and become positive. Anions gain electrons and become negative. Typical charges: groups 1 +1, 2 +2, 13 +3, 15 -3, 16 -2, 17 -1. The octet rule is a useful model with exceptions; H needs 2 electrons.

Total charge must be zero in an ionic formula. Al3+ and O2-: 2(+3)+3(-2)=0, so Al2O3. Mg2+ and PO4^3-: Mg3(PO4)2. Reduce ratios: Ca2+ and O2- give CaO, not Ca2O2.

Revision | Continued

Polyatomic ions to memorize

NH4+ ammonium; OH- hydroxide; NO3- nitrate; NO2- nitrite; SO4^2- sulfate; SO3^2- sulfite; CO3^2- carbonate; HCO3- hydrogen carbonate.

PO4^3- phosphate; HPO4^2- hydrogen phosphate; ClO- hypochlorite; ClO3- chlorate; ClO4- perchlorate; CN- cyanide; BrO3- bromate. Keep a polyatomic ion together: Al(NO3)3, (NH4)3PO4. Parentheses mean more than one whole ion.

Naming ionic compounds

Name the cation first, then the anion. Monatomic anions end in -ide. Keep polyatomic-ion names. A Roman numeral gives the variable-charge metal's charge, not the atom count.

FeCl3: iron(III) chloride. FeS: iron(II) sulfide. Mg(ClO3)2: magnesium chlorate. PbO2: lead(IV) oxide. CuCl: copper(I) chloride. Na2SO4: sodium sulfate. IUPAC standard names help scientists communicate chemical identities clearly.

Molecular names and hydrates

Prefixes: mono 1, di 2, tri 3, tetra 4, penta 5, hexa 6, hepta 7, octa 8, nona 9, deca 10. Usually omit mono on the first element. The second ends in -ide.

PCl5 phosphorus pentachloride; N2O5 dinitrogen pentoxide; CF4 carbon tetrafluoride; SO2 sulfur dioxide; NO nitrogen monoxide; S2Cl2 disulfur dichloride. CoCl2.6H2O: cobalt(II) chloride hexahydrate. Anhydrous CoCl2 has no water of crystallization.

Revision | Continued

Bonding and conductivity

Ionic bonding: electrostatic attraction between opposite ions. Electron transfer produces ions. Covalent bonding: sharing electron pairs. Ionic substances form lattices; simple molecular substances contain separate molecules.

Ionic solids: usually high melting, hard and brittle. Solid ions cannot move, so do not conduct. Molten or dissolved ions can move and conduct. Simple molecular substances usually have lower melting/boiling points and poor conductivity. Test melting point, molten conductivity and solution conductivity; solubility alone is not decisive.

Electronegativity and polarity

Electronegativity is the ability of a bonded atom to attract shared electrons. Difference = larger EN minus smaller EN. Higher EN atom is partially negative. Larger difference means a more polar bond.

Your PPT rule: 0 non-polar; between 0 and 1.7 polar; 1.7 or above predicts ionic. This is a guide: HF is polar covalent and CaS ionic despite borderline values. Your PPT labels C-H (0.4) weakly polar. Use the question's table and requested convention. H-F > H-Cl; O-H > C-H; C-N > N-N in polarity.

Lewis structures: count first

Total valence electrons: add atomic contributions; ADD electrons for negative charge, SUBTRACT for positive. Draw bonds, complete outer atoms, place remaining electrons, form multiple bonds if needed, then check totals and charges.

One bond = 2 electrons; double = 4; triple = 6. H2O: 8 total, 2 bonds, 2 lone pairs on O. NH3: 8 total, 3 bonds, 1 lone pair on N. CH4: 8 total, 4 bonds, no lone pair on C. Structural formulas omit lone-pair dots; Lewis structures include them.

Revision | Continued

Lewis structures: homework targets

I2: I-I, 3 lone pairs per I. CF4: 4 C-F bonds, 3 lone pairs per F. H2S: 2 S-H bonds, 2 lone pairs on S. PCl3 and NF3: 3 bonds and 1 lone pair on central atom; each halogen has 3 lone pairs.

C2H2: H-C triple-bond C-H; no carbon lone pairs. Ethanol: CH3-CH2-OH; ether: CH3-O-CH3; each O has 2 lone pairs. SO2 has resonance; one expanded-octet drawing has two S=O bonds and one S lone pair. CsCl is ionic: Cs+ and Cl-, not a covalent molecule.

Charged Lewis structures

H3O+: 8 electrons, 3 O-H bonds, 1 lone pair on O. ClO-: 14 electrons, Cl-O single bond, 3 lone pairs on each atom. HCO3-: 24 electrons, HO-C(=O)-O-, 2 equivalent contributors.

CO3^2-: 24 electrons; one C=O and two C-O- in each contributor; 3 equivalent contributors. NO3-: 24 electrons; N has +1 formal charge, two singly bonded O have -1 each. Put ions in brackets with the overall charge. Resonance contributors are models of delocalized bonding, not molecules switching back and forth.

Radiation and practical chemistry

Radioisotopes have unstable nuclei that decay and emit radiation. Alpha: helium nucleus, +2, stopped by paper. Beta-minus: electron, -1, reduced by aluminium. Gamma: electromagnetic radiation, neutral, strongly penetrating; thick lead or concrete attenuates it.

Uses in lessons: Am-241 smoke detectors; Co-60 food irradiation; C-14 dating once-living material; I-131 thyroid treatment; Tc-99m medical imaging. Flame tests: Li crimson, Na yellow, K lilac, Ca orange-red, Sr red, Cu blue-green. Colour comes from electrons releasing energy as light.

Revision | Continued

Nature of chemistry and applications

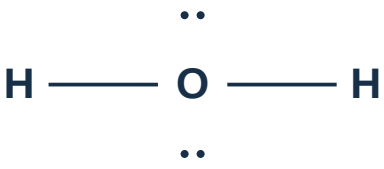
Matter has mass and occupies space. Chemistry studies matter, its properties and changes. Empirical knowledge comes from observation; theoretical knowledge uses explanations/models. Macroscopic means observable scale; microscopic means particles.

Science explains; technology applies. Green chemistry reduces hazardous substances, waste and energy use. IUPAC: International Union of Pure and Applied Chemistry. Sugar and sweeteners differ in structure and properties: compare taste, energy contribution, stability, cost and evidence-based suitability. Molecular shape matters as well as bond polarity.

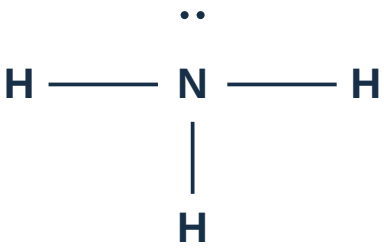
Lewis drawing practice

Each line is one shared pair; each pair of dots is a lone pair. Copy once, then redraw without looking.

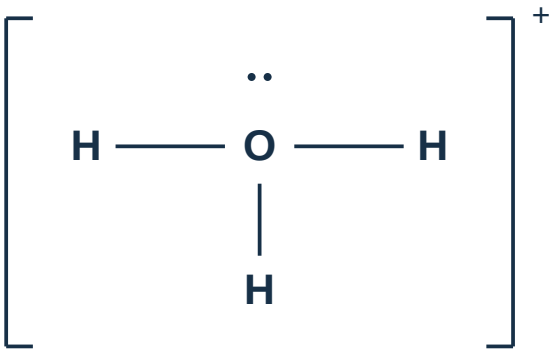
Water H2O: 8 electrons; 2 bonds and 2 oxygen lone pairs.



Ammonia NH3: 8 electrons; 3 bonds and 1 nitrogen lone pair.

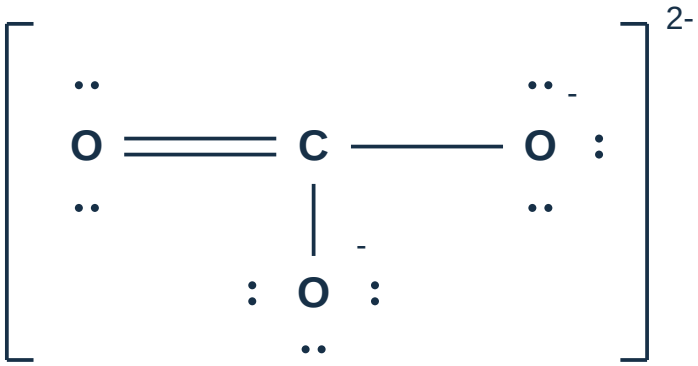


Hydronium H3O+: 8 electrons; 3 bonds and 1 oxygen lone pair.



Carbonate and homework corrections

One valid CO3^2- resonance contributor



Count 24 electrons: 8 in bonds, 16 in lone pairs. Carbon has an octet. The double-bond position has three equivalent choices. Actual bonding is delocalized; a Lewis drawing is a model.

Corrections to memorize

FeS is iron(II) sulfide, NOT sulfate. Sulfate contains SO4^2-.

Mg(ClO3)2 is magnesium chlorate, NOT chloride.

Your printed Q4(a) is PCl5: phosphorus pentachloride. The notebook copied PCl3.

CCl4 is carbon tetrachloride, NOT trichloride.

A full Lewis diagram needs lone pairs and charges, not just bond lines.

Your electronegativity PPT uses 0 / 1.7 boundaries, but actual bonding is not decided by a universal cutoff. HF is polar covalent; CaS is ionic. Use the supplied values and the convention explicitly requested.

Your Chapter 1 quiz Q22 is incomplete: it says "Na or" without the second element. Do not invent the missing element. Practise a complete comparison such as Na versus Mg (Mg has higher first ionization energy).

Additional homework answers | Naming

2.4 Q1: LiCl; K2S; FeCl2; Al2O3; Na2SO4; SnO2.

Q2: Name cation first; use Roman numeral for a variable-charge metal; name the anion, retaining a polyatomic name or using -ide for a monatomic anion.

Q3: magnesium chloride; cesium oxide; iron(II) sulfide; sodium phosphate; ammonium nitrate; aluminium sulfate; magnesium chlorate; lead(II) bromate; zinc hydrogen phosphate; sodium cyanide.

Q4: phosphorus pentachloride; dinitrogen pentoxide; carbon tetrafluoride; sulfur dioxide.

Q5: CoCl2.6H2O (hydrated); CoCl2 (anhydrous).

Q6: PCl3; CCl4; NO; S2Cl2.

Q7: potassium hydroxide; sodium nitrite; copper(I) chloride; sodium hydroxide; calcium carbonate.

Q8: Standardized IUPAC nomenclature prevents ambiguity and supports clear worldwide communication.

Additional homework answers | Polarity

Increasing electronegativity: Cs < K < Ca < Fe < Br < Cl < F. Use tabulated values for close or transition-metal comparisons.

More polar bonds: H-F, O-H, C-N.

N2 non-polar covalent; NH3 and H2O polar covalent bonds; FeO and MgCl2 ionic.

Homework approximate differences: Ca-S 1.6; H-F 1.8; P-H about 0; C-Cl 0.6; C-O 0.9; Li-Cl 2.2. The PPT uses slightly different rounded values (C-O 0.8). Use the values provided in each question.

Last-hour recall checklist

[ ] Can I define atomic number, isotope, ionization energy, electron affinity, electronegativity, cation, anion, ionic bond, covalent bond, polyatomic ion and radioisotope?

[ ] Can I calculate p, n, e, charge and an isotope weighted average?

[ ] Can I explain trends using shells, shielding and nuclear attraction?

[ ] Can I write neutral formulas with parentheses and name variable-charge metals?

[ ] Can I draw all lone pairs in H2O, NH3, H3O+ and carbonate?

[ ] Can I explain conductivity using mobile ions, not electrons flowing through salt?

[ ] Can I recognize alpha, beta and gamma and match flame-test colours?

[ ] Can I correct the homework errors without looking?

Source scope

Based on Grade_11_Chemistry_Chapters_1_2_Revision.docx, Grade_11Quiz.docx, ANSWER KEY.pdf, the supplied lesson PPT text on atomic structure, periodic table/trends, isotopes/radiation, ionic/molecular compounds, electronegativity, nomenclature, nature of chemistry and sweeteners, plus homework photos. Image-only slide details may not be captured by text extraction. This pack prioritizes the repeated assessable concepts.